

Study questions for oral exam on first course in Mathematical Statistics, MASA03

1 Probability Theory

1. Define the concept of a σ -algebra, and give two examples of such. Define the concepts of outcome space, elementary events and events.
2. Define union, intersection and complements of events. What does it mean that two events are disjoint?
3. Explain how one can, for an arbitrary collection $\mathcal{A}$ of subsets of Ω , construct a σ -algebra from $\mathcal{A}$, and prove that the constructed thing is a σ -algebra.
4. Give an example of two σ -algebras $\mathcal{F} \subseteq \mathcal{G}$. What does this inclusion intuitively mean?
5. Give the definition of a probability, so of a probability measure, $\mathbb{P}$.
6. Let $A, B \in \mathcal{F}$ where $\mathcal{F}$ is a σ -algebra. Prove:
 - (i) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
 - (ii) If $A \subseteq B$ then $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$.
7. Let $\{A_i\}_{i=1}^{\infty}$ be an increasing sequence $A_1 \subseteq A_2 \subseteq \dots$ of events. Let $A = \cup_{i=1}^{\infty} A_i$. Show that $\mathbb{P}(A) = \lim_{i \rightarrow \infty} \mathbb{P}(A_i)$.
8. Define conditional probabilities. Define independent events.
9. State and show the Bayes theorem, and the law of total probability.
10. Give the definition of a random variable X and of a (cumulative) distribution function $F(\cdot)$. Why is it important that $\{X \in A\} \in \mathcal{F}$ for each Borel $A \subseteq \mathbb{R}$ in order to define F ? Give an example of a function on the state space which is not a random variable.
11. Show that a distribution function F is right-continuous, increasing, and satisfies $\lim_{t \rightarrow \infty} F(t) = 1$ and $\lim_{t \rightarrow -\infty} F(t) = 0$.
12. When does a random variable have a density? Define the density of a random variable in this case. What types (three types) of random variables do you know?
13. Define the Riemann-Stieltjes integral $\int g(x) dF(x)$. Let X be a random variable with distribution function F . How can you express the expectation $\mathbb{E}X$ with the help of this integral? How can you simplify this formula when r.v. X is continuous? How can you simplify this formula when r.v. X is discrete?

14. How can you simplify the formula $\int g(x) dF(x)$ when the r.v. X is a mixture of a discrete and continuous distribution.
15. Define expectation, variance and standard deviation of a random variable.
16. Let $1\{A\}$ be the indicator function for the event A . Let $p = \mathbb{P}(A)$. What is $\mathbb{E}(1\{A\})$? What is $\text{Var}(1\{A\})$?
17. Define the Binomial distribution and describe how it arises. Derive the expressions for the probability mass function, from the description of how it arises. Derive the expressions for the expectation and the variance for this distribution.
18. Let $X = (X_1, \dots, X_n), Y = (Y_1, \dots, Y_n)$ be random vectors. Define $\mathbb{E}(X)$ and $\text{Cov}(X, Y)$. Show that $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$, where $a, b \in \mathbb{R}$ are some constants, in case $n = 1$.
19. If X is an n -dimensional random row vector, define $E(X)$ and $\text{Var}(X)$. If A is a $p \times n$ matrix (of nonrandom elements), a is a n -dimensional row vector (of nonrandom elements) and X is an n -dimensional random row vector, state the formulas for $\text{Var}(a + AX)$ and $E(a + AX)$.
20. Let X, Y be two random variables. Give two equivalent definitions for them being independent, using distribution functions or density functions. Show that the two definitions are equivalent, *at least in one direction*, in the case the random variables are continuous.
21. Show that X, Y independent implies X, Y uncorrelated. Give an example for when the opposite implication does not hold.
22. (i) Suppose that (X, Y) is a continuous bivariate random vector. Define the conditional probability density function for X given Y . Define $\mathbb{E}(X|Y)$. (ii) Suppose (X, Y) is discrete bivariate random vector. Define the conditional probability mass function for X given Y . Define $\mathbb{E}(X|Y)$. (iii) Show that $\mathbb{E}(X) = \mathbb{E}(\mathbb{E}(X|Y))$, for either (it is enough to prove the statement for one of them) the continuous or the discrete case.
23. Define the uniform $Un(0, \theta)$ distribution, where $\theta \in \mathbb{R}_+$. Derive the expectation and the variance for this distribution.
24. Define the exponential distribution $Exp(\theta)$, for $\theta \in \mathbb{R}_+$. Derive the expectation and the variance for this distribution.
25. Define the Normal distribution $\mathcal{N}(m, \sigma^2)$. Show that if $X \sim \mathcal{N}(0, 1)$ then $aX + b \sim \mathcal{N}(b, a^2)$, where $a, b \in \mathbb{R}$, $a \neq 0$.
26. State and prove the change of variable formula for the density of $Y = g(X)$ when X is a continuous random variable and g is continuously differentiable, and the derivative g' does not change sign.
27. Derive the convolution theorem for i) two discrete random variables, ii) two continuous random variable. How does the formula simplify when the random variables are independent?

28. State the “law of the unconscious statistician” for $Y = g(X)$. Prove the result for the case when X is a continuous r.v. and g is function that is strictly monotone (increasing or decreasing).
29. Suppose that $X \in \text{Bin}(n_1, p)$, $Y \in \text{Bin}(n_2, p)$ and X, Y are independent. Derive the distribution of $X + Y$.
30. Define the multivariate Gaussian distribution. You have seen two equivalent definitions, it is enough that you state one of them.
31. Define the four kinds of convergence of random variables that you have studied.
32. Give the statement of the weak/strong law of large numbers (LLN). Prove it in some cases (which you know). What is the use of the LLN (explicit and implicit use)?
33. Give a statement of the law of rare events. Prove it. What is the use of the law of rare events (explicit and implicit use)?
34. Give the statement of the central limit theorem (CLT). What is the use of the CLT (explicit and implicit use)?

2 Inference Theory

1. Let $x_1, \dots, x_n$ be an i.i.d. sample from a random variable X with distribution function $F_\theta \in \{F_\theta : \theta \in \Theta\}$. State the inference problem, based on the data $x_1, \dots, x_n$. Define the concepts of parametric, non-parametric and semi-parametric inference problems. Define a statistic.
2. Which are the three general ways to make inference about a parameter? Exemplify these for the situation x an observation of X which is $\text{Bin}(25, \theta)$ -distributed.
3. Define the empirical distribution F_n based on a sample $x_1, \dots, x_n$. Viewing F_n as a random element, what is the distribution of $nF_n(x)$? Derive from this property, $\mathbb{E}(F_n(x))$ and $\text{Var}(F_n(x))$.
4. Let $x_1, \dots, x_n$ be i.i.d. observations of X with unknown distribution function $F \in \{F : F \text{ distribution function on } \mathbb{R}\}$. For $g : \mathbb{R} \rightarrow \mathbb{R}$ a function, describe $\eta := \mathbb{E}(g(X))$ as a functional of F . Explain for which g this definition make sense. Define the plug-in estimator of η . Apply the definition on the estimation of $\eta = \mathbb{E}(X^3)$, and on $\mathbb{E}(X \cdot 1\{X \in [0, 1]\})$.
5. What is an unbiased estimator? Define the mean square error (m.s.e.) of an estimator. Prove the partition of the m.s.e. into it's bias and variance parts.
6. Define consistency of an estimator, in quadratic mean, almost surely and in probability. Show that consistency in quadratic mean implies consistency in probability.
7. Show that the plug-in estimator of a functional of the form $\theta = \int g(x)dF(x)$ is consistent in quadratic mean.

8. Let $x_1, \dots, x_n$ be a sample from a distribution F_θ , with $\theta \in \Theta \subset \mathbb{R}^s$, with $s < \infty$. Describe the maximum likelihood method for estimating a parameter θ .
9. Let $x_1, \dots, x_n$ be a sample from $N(m, \sigma^2)$, with m and σ^2 unknown. Derive the maximum likelihood estimators of μ and σ^2 .
10. Let x be a sample from $\text{Bin}(n, p)$ with p unknown. Derive the maximum likelihood estimator of p .
11. Let $x_1, \dots, x_n$ be a sample from X which is $\text{Un}[\theta_1, \theta_2]$ -distributed, with $\theta = (\theta_1, \theta_2)$ unknown. Find the maximum likelihood estimators of θ .
12. Let $x_1, \dots, x_n$ be a sample from a distribution F_θ , with $\theta \in \Theta \subset \mathbb{R}$ unknown. Describe the least-squares method (ls) for estimating a parameter θ .
13. Derive the least squares estimators for the parameters m and p in questions 9 and 10.
14. Describe the idea behind a confidence interval for an unknown one-dimensional parameter θ in a continuous distribution, with a certain confidence grade. Describe the method with pivot function or pivot variable for constructing a confidence interval.
15. Describe the idea underlying a statistical test. Define the concepts null- and alternative hypothesis, test statistics, critical region and significance level. Describe the method with pivot function or pivot variable for constructing tests for a one-dimensional parameter in a continuous distribution. Define the power function for a given test.
16. Describe the p-value method for testing.
17. Suppose that θ is the unknown parameter in a *continuous* distribution. Can you use both the p-value method and the method with critical regions for testing? Suppose that θ is the unknown parameter in a *discrete* distribution. Can you use both the p-value method and the method with critical regions for testing?
18. Let $x_1, \dots, x_n$ be a sample from $N(m, \sigma^2)$, with m unknown, σ^2 known. Derive a 95% two-sided confidence interval m . Derive a test on significance level 5% for the hypotheses $H_0 : m \geq 1$, $H_1 : m < 1$.
19. (ctd.) Derive the power function of the test in the previous question.
20. Suppose that $x_1, \dots, x_n$ is sample from $N(m, \sigma^2)$, with both m and σ^2 unknown. Derive a 99% one-sided lower bounded confidence interval for m . Derive a test on significance level 1% for the hypotheses $H_0 : m = 2$, $H_1 : m \neq 2$.
21. Suppose that $\theta = (\theta_1, \dots, \theta_k)$ is a k -dimensional parameter and suppose that somehow you have derived, or you have been given, k confidence intervals $I_1, \dots, I_k$, with each I_j a confidence interval for θ_j with the same confidence grade $1 - \alpha$. Is $I = I_1 \times \dots \times I_k$ a (simultaneous) confidence set for θ , with confidence grade $1 - \alpha$? Why not? Give a lower bound on the confidence grade for the simultaneous confidence set I .
22. Describe the problem with repeated tests when testing for a parameter $\theta = (\theta_1, \dots, \theta_k)$. Describe the Holm-Bonferroni correction method for multiple testing.

23. Describe the confidence method for doing tests. Give an example for when the confidence method does not give the same test as a standard (T, C) test.
24. If X is a continuous random variable with density, why can you not use the plug-in approach to estimating $\eta = T(F) = f(x_0)$, for a fixed x_0 ? Define the kernel estimator of $\eta = f(x_0)$. Why is the kernel estimator preferable to the histogram?
25. How do you construct non-parametric one-sample tests with the use of the empirical distribution function? How do you construct confidence bands for F ? How do you construct a k -sample test?
26. Suppose that θ is the linear functional in problem 7, and let $\hat{\theta}_n$ be the plug-in estimator. Suppose that $\text{Var}(g(X)) = \sigma^2 < \infty$ is known. What is the limit distribution of $\hat{\theta}_n$? Derive an approximate 95% confidence interval for $\mathbb{E}(g(X))$. What happens when σ^2 is not known?
27. State the continuous mapping theorem and Slutsky's theorem. Give one situation in which are they used?
28. Suppose that $(x_{1i} \dots, x_{pi}, y_i), i = 1, \dots, n$ are observations in the regression problem

$$y_i = \theta_0 + \theta_1(x_{1i} - \bar{x}_{1\cdot}) + \dots + \theta_p(x_{pi} - \bar{x}_{p\cdot}) + \epsilon_i,$$

- with $\epsilon_i \in N(0, \sigma^2)$ i.i.d. random variable. Write the regression problem on matrix form. Derive the least squares estimator $\hat{\theta}$ of the vector $\theta = (\theta_0, \dots, \theta_p)$. Show that $\hat{\theta}$ is unbiased. Derive the covariance matrix of $\hat{\theta}$. What is the distribution of $\hat{\theta}$?
29. (ctd.) Explain how you can use this knowledge to make tests and confidence intervals for the θ -parameters in the previous question, when σ^2 is known. If $a_1, \dots, a_p$ are known and $\eta = a_1\theta_1 + \dots + a_p\theta_p$ a parameter of interest, how do you make tests and construct confidence intervals for η ? What property of the multivariate Gaussian distribution do you use for this?
 30. (ctd.) Derive the maximum likelihood estimator for θ in the previous question. Derive the maximum likelihood estimator of σ^2 . Explain how you can make tests and confidence intervals for θ when σ^2 is unknown.